

Capacity is not Calibration: A Pre-Registered Distribution-Aware Benchmark of Single-Cell Perturbation Predictors

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Abstract

We introduce ConfPert, a benchmark that evaluates 12 single-cell perturbation predictors with finite-sample conformal coverage as the primary outcome rather than mean-prediction accuracy. The benchmark covers 8 publicly available Perturb-seq and chemical-perturbation datasets and ships with a machine-verifiable YAML pre-registration that a command-line verifier refuses to evaluate when the locked hypotheses, thresholds, or BH-FDR corrections do not match the results file. Across 651 locked (predictor, dataset, α , discrepancy) cells, the pre-registered family-level Kruskal-Wallis test (H_3) rejects equality of calibration deviation between the baseline family (F_1 , mean / bilinear-ridge / noisy-mean; $\bar{d} = 0.078$, $n = 25$) and the transformer-foundation-model family (F_3 , scGPT / scFoundation / Geneformer in frozen-encoder embedding mode; $\bar{d} = 0.154$, $n = 25$) at $p_{KW} = 1.7 \times 10^{-7}$ with Cliff’s $\delta = 0.80$. The hypotheses on per-dataset capacity (H_1) and cell-line context (H_2) also pass; the hypotheses on training-corpus size (H_{1b}) and corpus-diversity (H_{2b}) fail. The result reproduces the perturbation prediction skepticism of Ahlmann-Eltze et al. [2025], Csendes et al. [2025] under a coverage criterion. We release the library on PyPI (**confpert** v0.2.0rc2), the pre-registration template, and a Codabench leaderboard for community submissions.

1 Introduction

Predicting single-cell responses to genetic or chemical perturbations is the headline application of single-cell foundation models. Recent benchmarks report that simple linear baselines match or exceed deep models on standard transcriptomic prediction metrics Ahlmann-Eltze et al. [2025], Csendes et al. [2025], Kedzierska et al. [2025], but the metrics in these benchmarks are still primarily measures of mean-vector accuracy (MSE, Pearson, top- k DEG recovery). For downstream uses such as cell-population sampling or risk-aware drug response, coverage of the predicted distribution – not mean accuracy – is the actionable property.

We evaluate predictors by per-population conformal coverage. Six discrepancy scores – Kolmogorov-Smirnov, 1-Wasserstein, energy, MMD-RBF, bimodality mismatch, and variance-ratio deviation – are wrapped in finite-sample conformal heads Vovk [2012], Tibshirani et al. [2019], Romano et al. [2019], Izbicki et al. [2022], Cauchois et al. [2021], Bates et al. [2021], and we report calibration deviation $|\hat{\alpha} - \alpha|$ as the primary outcome. The pipeline is locked in a YAML pre-registration before any first run; a CLI verifier (**confpert prereg verify**) refuses to evaluate when the YAML and the results file disagree. Pre-registration in machine learning has been argued for Hofman et al. [2023], Pineau et al. [2020] but remains rare; the closest mechanically enforced precedents we are aware of are competition-style leaderboards

(e.g. Kaggle), not hypothesis-level locks. [CITATION NEEDED] for a single-cell-perturbation benchmark with a comparable mechanically enforced lock.

Empirical finding. The pre-registered Kruskal-Wallis test (H_3) rejects equality of calibration deviation across the three predictor families ($p_{\text{KW}} = 1.7 \times 10^{-7}$, Cliff’s $\delta = 0.80$). The baseline family (F_1) is the best-calibrated; the foundation-model family (F_3) is the worst. Per-dataset capacity (H_1 , 4 of 8 datasets at $\rho > 0.5$, $p < 0.05$) and cell-line context (H_2 , two-way ANOVA f - $p = 0.0064$, $\eta^2 = 0.120$) also pass; corpus-size and corpus-diversity hypotheses (H_{1b} , H_{2b}) fail.

Contributions.

- A benchmark of 12 published predictors across 8 public single-cell perturbation datasets with within-perturbation and cross-cell-line splits; 651 locked result cells.
- A machine-verifiable YAML pre-registration with a CLI verifier that refuses to run when the locked YAML and the results file disagree.
- An ablation panel that re-runs the same hypothesis tests under counterfactual analyst-side relaxations, quantifying the threshold-shopping bias avoided by the lock.
- A pip-installable library, datasheet and model card, and a Codabench leaderboard.

2 Related work

Conformal prediction for multivariate and generative outputs. Conformal prediction has been extended to the multivariate and generative-model setting in a recent cluster of methods Meyer et al. [2026], Zheng et al. [2024], Thurin et al. [2025], Ndiaye [2025], Dheur et al. [2025], Yang et al. [2026], Braun et al. [2025], Klein et al. [2025], including a perturbation-data application Asiaee et al. [2026]. ConfPert sits downstream of this literature and operationalises six standard per-population discrepancies inside finite-sample conformal heads.

Single-cell foundation models and perturbation prediction benchmarks. Several recent benchmarks evaluate single-cell foundation models on perturbation tasks. Ahlmann-Eltze et al. [2025] compare five foundation models and two other deep learning models against deliberately simple baselines on Perturb-seq and report that none outperformed the baselines. Csendes et al. [2025] replicate the finding for scGPT and scFoundation against a mean-of-training-examples baseline. Kedzierska et al. [2025] show that scGPT and Geneformer can underperform simpler methods in zero-shot deployments. Boiarsky et al. [2024] reach a similar conclusion under a deeper single-cell foundation-model evaluation. Li et al. [2025] surveys the largest pool of perturbation predictors to date but reports accuracy-style scores (MSE, mean Pearson) rather than calibration coverage. ConfPert’s incremental contribution is to add a coverage criterion, a finite-sample conformal head per discrepancy, and a mechanically enforced pre-registration.

Pre-registration in machine learning. Pre-registration is established in social and biomedical research Nosek et al. [2018] and has been argued for in predictive modelling Hofman et al. [2023], Pineau et al. [2020]. The ConfPert verifier extends this prior work by refusing to evaluate when the locked YAML and the on-disk results file disagree.

3 Benchmark design

3.1 Predictors

We benchmark 12 predictors across three pre-registered family levels. F_1 baselines: mean (single Perturb-seq mean per perturbation), ahlmann_bilinear_ridge Ahlmann-Eltze et al. [2025]

(additive model with ridge regularisation, $K = 10$, $\lambda = 0.1$), and `noisy_mean` (mean with isotropic noise). F_2 latent-delta: `scGen` Lotfollahi et al. [2019], `biolord` Piran et al. [2024], `CPA` Lotfollahi et al. [2023], `sVAE+` Bereket and Karaletsos [2023], and `STATE` Adduri et al. [2025]. F_3 transformer: `GEARS` Roohani et al. [2024], `scGPT` Cui et al. [2024], `scFoundation` Hao et al. [2024], and `Geneformer` Theodoris et al. [2023].

Foundation models run in frozen-encoder mode. A ridge regression head (`sklearn.linear_model.Ridge`, $\alpha_{\text{ridge}} = 1.0$) is fitted on the train fold of cell-level encoder embeddings against the `log1p HVG` expression target; the per-gene sampling variance used in the predictive distribution is computed only on training-fold rows via `confpert.heavyweight_helpers.safe_train_variance`, which raises if the supplied train and test row indices overlap. This test-fold isolation is enforced as code so that a future contributor cannot silently re-introduce the leak. Full fine-tuning of the foundation models is left to future work; the present results should be read as upper bounds on the F_3 family under transfer-learning use, not as a verdict on the architectures.

3.2 Datasets

We use 8 datasets covered by at least one filled result row: `Norman 2019` Norman et al. [2019], `Replogle K562` and `RPE1 essential` Replogle et al. [2022], `Adamson 2016` Adamson et al. [2016], `Tahoe-100M subset` Zhang et al. [2025], `Frangieh` Frangieh et al. [2021], `Schmidt` Schmidt et al. [2022], `Datlinger` Datlinger et al. [2017], and `Lara-Astiaso 2023` ex-vivo mouse `Perturb-seq` [NEEDS USER INPUT] for full in-text citation of the `Lara-Astiaso 2023` Nature Genetics primary source. `McFaline-Figueroa` `McFaline-Figueroa` et al. [2024] and `CellFM` Zeng et al. [2024] are documented in the pre-registration but report as N/A in the result table (sample-meta hash-table demultiplexing for `McFaline` and absence of a usable HF mirror for `CellFM` as of 2026-05-25). The mouse-organism slot was originally pre-registered for “Walker et al. Nature Methods 2022 BMDC Perturb-seq”; that citation could not be cross-verified, and the slot was substituted with `Lara-Astiaso 2023` on 2026-05-25 with an explicit amendment block in the YAML lock (option B of the three documented resolution paths in `confpert/data/walker.py`).

3.3 Splits

Within-perturbation: each perturbation’s cells are partitioned 50/25/25 into train/calibration/test by seed 42; control cells are split proportionally. Held-out-perturbation: 50/50 perturbation-level split by the same seed. Cross-cell-line: train on `Replogle K562`, evaluate on `Replogle RPE1` with gene-name intersection. Held-out-organism (train `Lara-Astiaso`, test `Norman`), held-out-modality (`Frangieh RNA` train, `Frangieh protein` test), and held-out-perturbation-type (`Replogle K562` train, `Tahoe chemical` test) splits are pre-registered but the held-out-modality and held-out-perturbation-type cells report as [NEEDS USER INPUT] in the present run (those split-specific pipelines were not executed within the Modal budget).

3.4 Discrepancies and conformal heads

Six per-population discrepancies: Kolmogorov-Smirnov on per-gene marginals, 1-Wasserstein on per-gene marginals, energy distance Székely and Rizzo [2013], MMD with an RBF kernel Romano et al. [2019], Izibicki et al. [2022], bimodality mismatch (KDE-based per-gene bimodality divergence), and variance-ratio deviation ($\text{var}(\text{pred}) / \text{var}(\text{obs}) - 1$). Each discrepancy is wrapped in a per-perturbation conformal head Vovk [2012], Tibshirani et al. [2019], Romano et al. [2019]; subgroup-conditional Cauchois et al. [2021] and RCPS Bates et al. [2021] heads are implemented in the library but evaluated as [NEEDS USER INPUT] in this paper.

3.5 Pre-registration protocol

The hypotheses, thresholds, BH-FDR corrections, and disposition rules are committed to `paper_neurips_dnb_2` before any first run; the file’s git commit SHA and SHA-256 hash are written into the YAML’s lock block. `confpert prereg verify` refuses to run if the on-disk YAML hash differs from the locked hash, if the git commit SHA on the YAML differs from the locked SHA, or if the `results_json_max_mtime` field is exceeded by the results file’s mtime. Re-emitting hashes (`confpert prereg emit-hashes`) is required after any intentional amendment and is itself recorded by a git commit.

4 Materials and Methods

Software environment. Python 3.11 (CPU) and 3.10 (CPA / scvi-tools subset). Key version pins: `torch 2.4.1` (`torch 2.3.1` in the scGPT image to keep `torchtext 0.18.0` ABI-compatible), `scvi-tools<1.4` for scGen / biolord, `cell-gears 0.1.2` for GEARS, `scgpt 0.2.4`, `transformers>=4.40,<4.46`. The full image graph (one per heavyweight predictor family) is defined in `scripts/modal_launch.py`. [NEEDS USER INPUT] for the exact Docker image SHA when frozen for camera-ready.

Hardware. Heavyweight predictor runs were executed on Modal A100 GPUs (one A100 per (predictor, dataset) call, 24–32 GB host RAM); lightweight predictor runs on a local M-series Apple Silicon CPU. The Modal budget cap was \$3000; the actual total spend on the locked sweep was [NEEDS USER INPUT] for the camera-ready figure.

Datasets and preprocessing. The eight datasets and their source URLs are listed in the YAML (`k1_v2.datasets`) and reproduced in Appendix A. All datasets are loaded by the common pipeline (`confpert.data._load_perturb_ds`): drop cells with fewer than 200 UMI; drop genes detected in fewer than 5 cells; `scanpy.pp.normalize_total(target_sum=1e4)`; `scanpy.pp.log1p`; `scanpy.pp.highly_variable_genes` with $n_{\text{top}} = 512$. The top 50 perturbations by cell count are retained (29 for Datlinger). Random seed is 42 throughout.

Predictor fitting. F_1 predictors are fit on the train fold and noise variants A–D sweep isotropic, per-gene marginal, and full-covariance Gaussian noise. F_2 and F_3 predictors are fit per the upstream package defaults; all per-predictor configurations are stored in `baselines/results.json:rows[*].conf`. Foundation models pass through `safe_train_variance` and `sample_from_train_distribution` (`src/confpert/heavyweight_helpers.py`); both refuse mis-split inputs.

Discrepancy and conformal-head computation. For each held-out perturbation p , we draw a predicted population $\hat{X}_p$ of size matching the observed cell count, compute each of the six discrepancies $s_i(\hat{X}_p, X_p)$, and run a finite-sample per-perturbation conformal calibration on the train half of the perturbations. The resulting threshold τ_α is applied to the held-out perturbations and we report achieved coverage $\hat{\alpha} = \Pr[s_i \leq \tau_\alpha]$.

Hypothesis tests. H_1 : Spearman ρ between $\log_{10}(\text{param count})$ and mean calibration deviation per dataset; pass if $\rho > 0.5$, $p < 0.05$ (permutation, $n = 10000$), in ≥ 3 of 4 main datasets. H_{1b} : Spearman with training-corpus size, $\rho < -0.5$, same ≥ 3 of 4 rule. H_2 : Two-way Type-II ANOVA on calibration deviation with factors (cell-line context, log param count); pass if main-effect f - $p < 0.0167$ (Bonferroni on 3 family tests) and $\eta^2 > 0.10$. H_{2b} : Spearman with log corpus diversity, $\rho < -0.5$, $p < 0.0167$, ≥ 7 of 10 datasets. H_3 : Kruskal-Wallis across the three families, $p < 0.0167$ and Cliff’s $\delta > 0.50$ between best and worst.

Bootstrap CIs and power. Wilson 95% binomial intervals are computed per (predictor, dataset, score, α) cell; per-cell counts and intervals are reported in `phase2d_report_phase2.json:bootstrap_ci_table`. Power estimates use Monte Carlo with $n_{\text{sims}} = 2000$ for the Spearman / Kruskal tests and `statsmodels.stats.power.FTestAnovaPower` for the ANOVA.

Reproduction.

`git checkout <commit-SHA>`

```

pip install -e .
python -m confpert.cli phase2d \
  --prereg paper_neurips_dnb_2026/preregistration_v2.yaml \
  --results baselines/results.json \
  --out paper_neurips_dnb_2026/phase2d_report_phase2.json

```

5 Results

5.1 K1 sweep

The K1 sweep produced 651 cells across 12 predictors, 8 datasets, 3 target coverage levels ($\alpha \in \{0.05, 0.10, 0.20\}$), and 6 discrepancies. Mean calibration deviation per family is $\bar{d}_{F_1} = 0.078$ ($n = 25$ cells), $\bar{d}_{F_2} = 0.123$ ($n = 26$), and $\bar{d}_{F_3} = 0.154$ ($n = 25$). The per-discrepancy hardness rank (mean deviation across all predictor-dataset pairs, ascending) is $\text{KS} < \text{bimodality_mismatch} < \text{MMD-RBF} < W_1 < \text{energy} < \text{variance_ratio_dev}$. The cross-dataset heatmap is in Figure ?? and the calibration-vs-capacity Pareto is in Figure ??; full per-cell results with Wilson 95% CIs are in Appendix C and the JSON sidecar.

5.2 K2 v2 hypothesis tests

The five pre-registered hypotheses resolved as follows (Table 1). H_1 *capacity* passes: 4 of 8 datasets (Norman, Frangieh, Replogle RPE1, Tahoe) meet the per-dataset $\rho > 0.5$, $p < 0.05$ rule. H_{1b} *data* fails: per-dataset Spearman is degenerate because many predictors share corpus size. H_2 *cell-line context* passes: two-way ANOVA f - $p = 0.0064$ (< 0.0167) and $\eta^2 = 0.120$ (> 0.10). H_{2b} *corpus diversity* fails: $\rho > 0$ on every dataset (opposite sign of the hypothesised $\rho < -0.5$). H_3 *architecture family* passes: Kruskal-Wallis $p = 1.7 \times 10^{-7}$ with Cliff's $\delta = 0.798$ between F_3 transformer ($\bar{d} = 0.154$, $n = 25$) and F_1 baselines ($\bar{d} = 0.078$, $n = 25$). The disposition emitted by the verifier is `double_pass_no_corpus`.

Table 1: K2 v2 hypothesis dispositions (Phase 2D, verifier output). f - p is the two-way Type-II ANOVA main-effect p -value for the cell-line-context factor; p_{KW} is the Kruskal-Wallis p -value across the three families.

Hypothesis	Status	Observed
H_1 capacity	PASS	4 of 8 datasets at $\rho > 0.5$, $p < 0.05$
H_{1b} data	FAIL	0 of 8 datasets (ρ undefined per dataset)
H_2 cell-line	PASS	f - $p = 0.0064$, $\eta^2 = 0.120$
H_{2b} corpus	FAIL	per-dataset $\rho > 0$ on every dataset
H_3 family	PASS	$p_{\text{KW}} = 1.7 \times 10^{-7}$, Cliff's $\delta = 0.80$

5.3 Ablation panel

The ablation panel re-evaluates the same hypotheses under counterfactual analyst-side threshold relaxations. On a frozen Phase 1 snapshot (`tests/fixtures/phase1_results.json`), the locked $p < 0.05$ threshold passes only one dataset (Replogle RPE1) for H_1 , but the relaxation $\{p < 0.10 \text{ AND } n_{\text{pass}} \geq 2\}$ admits a second passer (Tahoe, $p = 0.066$) and flips the disposition. Under the locked thresholds, the flip never occurs. On the Phase 2D full data, H_1 passes under the locked rule (4 of 8 datasets); the Phase 1 flip case is preserved as evidence of the bias the lock prevents.

K3 PRISM downstream utility. Of the 445 ConfPert-calibrated PRISM signatures Corsello et al. [2020] that pass BH-FDR, the top-25 ranked by mean coverage deviation include 14 (56%) with

at least one annotated drug-target match in DrugBank or ChEMBL. The full match table is at `results/k3_tahoe_predictor_derived_top25.json`.

6 Discussion

Observation 1: foundation models are the worst-calibrated family. F_3 transformer predictors used in frozen-encoder mode produce predictive populations with the largest calibration deviation across all six discrepancies. A plausible mechanism is that the frozen encoder embeddings, even when followed by a linear head fitted on the train fold, lose the per-gene variance structure required for matching the full Perturb-seq distribution; their advantage on supervised embedding-to-label tasks does not transfer to per-population sampling. The finding is consistent with Ahlmann-Eltze et al. [2025], Csentesi et al. [2025], and Kedzierska et al. [2025], all of whom report that foundation models lose to a mean baseline on perturbation prediction under their respective accuracy metrics. The present paper adds the coverage criterion to that picture but does *not* settle the full-fine-tune question: we evaluate the transfer-learning use mode (frozen encoder + ridge head) that practitioners adopt when downstream labels are scarce, and the result should be read with that scope in mind.

Observation 2: cell-line context is a real covariate. The two-way ANOVA $\eta^2 = 0.120$ on cell-line context exceeds the pre-registered threshold, indicating that calibration deviation systematically differs between K562-derived and non-K562 contexts. A plausible mechanism is that the training corpora for the F2 and F3 predictors are heavily K562-weighted, leading to better calibration on in-distribution K562 splits than on the more diverse cell lines covered in Phase 2 (Frangieh melanoma, Schmidt T cells, Datlinger Jurkat).

Observation 3: corpus diversity does not help calibration. H_{2b} predicted a negative correlation between log corpus diversity and mean calibration deviation; the observed sign is positive on every dataset. One reading is that predictors trained on broader corpora may overfit to corpus-level structure that does not transfer to held-out perturbations; another reading is that our corpus-diversity index (`predictor.corpus_diversity_index` in the YAML) is a poor proxy. A future experiment would re-fit each predictor on subsamples of a single source corpus to dissociate diversity from total cell count.

Limitations. *Marginal, not conditional coverage.* Distribution-free conditional coverage is impossible Vovk [2012]; ConfPert reports marginal coverage within head levels. *Cancer-corpus dominance.* 7 of 8 evaluated datasets are derived from cancer or transformed cell lines (Lara-Astiaso primary cells is the exception); the primary-cell shift remains a known confound. *Heavy-weight predictor restriction.* Per pre-reg §14.5, the F3 family was evaluated on the 6 heavyweight-allowed datasets only; McFaline and Lara-Astiaso lack F3 cells. *Frozen-encoder mode.* The F3 family is evaluated in transfer-learning use, not full fine-tune; a full `scgpt.Trainer.train_perturb` implementation is left to future work and is expected to narrow but probably not close the gap. *Power.* H_1 , H_{1b} , H_{2b} are individually underpowered at their target effect sizes (Appendix E); the headline H3 disposition is well-powered.

Concrete future work. (1) Re-evaluate F_3 under full perturbation fine-tune (`scgpt.Trainer.train_perturb`) on the same six datasets and report the calibration gap that closes; an effect-size threshold of $\Delta\bar{d} \geq 0.05$ would meaningfully move the H_3 disposition. (2) Run the same pre-registered tests on the Replogle Genome-Wide Perturb-seq (with appropriate compute) to test whether the H3 PASS replicates on a larger single-dataset sample. (3) Add a CellFM wrapper once a working PyTorch port appears (biomed-AI/cellfm-800M returns 404 as of 2026-05-25) and re-test

H_3 . (4) Replace the corpus-diversity index with a TF-IDF-style cell-type-coverage measure and re-test H_{2b} .

7 Library and reproducibility

The benchmark ships as `confpert` v0.2.0rc2 on PyPI with 126 passing unit tests; a Cell-Eval plug-in covers 5 of 6 discrepancies absent from the upstream Arc Institute registry; a Codabench leaderboard at `paper_neurips_dnb_2026/codabench/` accepts community submissions. The datasheet (Geburu et al. 2021; `paper_neurips_dnb_2026/datasheet.md`) and model card (Mitchell et al. 2019; `model_card.md`) follow the templates of their respective sources. The Modal volume `causeflow-artifacts` holds frozen h5ads and per-job result JSONs keyed to the locked YAML SHA.

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- Ze Zheng et al. Generative conformal prediction with vectorized non-conformity scores. *arXiv preprint*, 2024. Sample-from-generative-model angle; medium scoop overlap with ConfPert.

A Datasheet for ConfPert

The full datasheet (Gebru et al. [2021] template) is at `paper_neurips_dnb_2026/datasheet.md` and is inlined verbatim into the camera-ready appendix. It documents motivation, composition (per-dataset n_{cells} , perturbation modality, organism, license), collection process (URLs in §3.2), preprocessing ($n_{\text{hvg}} = 512$ HVG selection, `log1p` normalisation), uses and tasks, and distribution and maintenance.

B Model card for the confpert library

The model card is at `paper_neurips_dnb_2026/model_card.md` and is inlined verbatim into the camera-ready appendix. It covers model details (six conformal heads, one head level), intended use (calibration coverage benchmarks), factors (predictor family, dataset context), metrics (six discrepancies), evaluation data (the eight evaluated datasets), and ethical considerations (cancer-corpus dominance, primary-cell underrepresentation).

C Full K1 results with Wilson 95% CIs

The per-cell K1 results (651 rows + Wilson 95% intervals) are stored as a JSON sidecar at `paper_neurips_dnb_2026/phase2d_report_phase2.json` under the key `bootstrap_ci_table`. Each row is keyed by (predictor, dataset, score, α) and reports the held-out perturbation count, observed coverage $\hat{p}$, and CI bounds. The camera-ready will inline a 30-row summary leaderboard; the full table remains in the JSON to keep the PDF under the 9-page main-body cap.

D Pre-registration verifier output

CLI reproduction:

```
python -m confpert.cli phase2d \
  --prereg paper_neurips_dnb_2026/preregistration_v2.yaml \
  --results baselines/results.json \
  --out paper_neurips_dnb_2026/phase2d_report_phase2.json
```

Verbatim disposition table is in Table 1; K2 v2 disposition: `double_pass_no_corpus`.

E Power analysis (K1 + K2 v2)

Pre-registered power computations (Phase 2D `power_report`):

Test	n	target effect	estimated power
H_1 per-dataset Spearman	14	$\rho = 0.5$	0.41 (underpowered)
H_{1b} cross-dataset Spearman	10	$\rho = -0.5$	0.25 (underpowered)
H_2 two-way ANOVA	120	$\eta^2 = 0.10$	0.81
H_{2b} per-dataset Spearman	14	$\rho = -0.5$	0.26 (underpowered)
H_3 Kruskal-Wallis (3 families)	120	$\delta = 0.5$	0.92

H_1 , H_{1b} , and H_{2b} are individually underpowered at their target effect sizes; the cross-dataset meta-analysis recovers usable power for H_1 in the headline disposition because the observed within-dataset effect sizes were larger than the target ($\rho \in [0.59, 0.71]$ on the passing datasets). H_2 and H_3 are independently well-powered.